Atharv Sonwane

Research Fellow, Microsoft Research

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Research Interests: *Reasoning + Learning, Robotics, Software engineering*

Education

Cornell University *2024 -*
PhD, Computer Science | Advisor: Kevin Ellis

Birla Institute of Technology and Science, Pilani *2018 - 2022*
Bachelor in Engineering, Computer Science.

Experience

Microsoft Research Montreal *Summer 2025*
Research Internship | Advisors: Eric Yuan and Alessandro Sordoni

- **BugPilot**: Training language models for agentic SWE tasks using synthetically generated bugs.

Microsoft Research India *2022 - 2024*
Research Fellow | Advisors: Dr. Aditya Kanade and Dr. Sriram Rajamani

- **MASAI**: A modular agentic framework for software engineering with 28.3% on SWE-bench-lite (SOTA at the time).
- **CodePlan**: Repository-level code editing with program analysis as interface for LLMs. **FSE '24**
- LLM self-evaluation for fixing code-quality issues. **FSE '24**

Robot Vision and Learning Lab, University of Toronto *2021*
Research Intern | Advisor: Dr. Florian Shkurti

- Designed and implemented lazy search + learning from experience for task and motion planning (TAMP) on tabletop robotics tasks. **ICRA '23**. Best Paper @ Long Horizon Planning Workshop @ CoRL 2022

TCS Research & Innovation *2021*
Research Intern | Advisor: Dr. Gautam Shroff

- Synthesising neural programs to represent analogical concepts. Student Poster **AAAI '21**, **NeSy '21**.

APP Centre for Artificial Intelligence Research *2021*
Undergraduate Researcher | Advisor: Prof Ashwin Srinivasan

- Solving visual reasoning (Bongard) problems using program synthesis (Dreamcoder) for representation and Inductive Logic Programming (ILP) for concept identification. **AAIP '21**

Publications / Preprints

OmniCode: A Benchmark for Evaluating Software Development Agents
Atharv Sonwane, *Eng-Shen Tu, Wei-Chung Lu, Claas Beger, Carter Larsen, Debjit Dhar, Simon Alford, Rachel Chen, Ronit Pattanayak, Tuan Anh Dang, Guohao Chen, Gloria Geng, Kevin Ellis, Saikat Dutta*
Findings of the Association for Computational Linguistics (**ACL Findings**) 2026. Link.

BugPilot: Complex Bug Generation for Efficient Learning of SWE Skills
Atharv Sonwane, *Isadora White, Hyunji Lee, Matheus Pereira, Lucas Caccia, Minseon Kim, Zhengyan Shi, Chinmay Singh, Alessandro Sordoni, Marc-Alexandre Côté, Xingdi Yuan*
arXiv preprint 2025. Link.

MASAI: Modular Architecture for Software-engineering AI Agents
Atharv Sonwane*, *Daman Arora*, Nalin Wadhwa*, Abhav Mehrotra, Saiteja Utpala, Ramakrishna Bairi, Aditya Kanade, Nagarajan Natarajan*
Workshop on Open-World Agents @ NeurIPS 2024 Link.

CodePlan: Repository-level Coding using LLMs and Planning

Ramakrishna Bairi, ***Atharv Sonwane****, Aditya Kanade, Vageesh D C, Arun Iyer, Suresh Parthasarathy, Sriram Rajamani, B. Ashok, Shashank Shet*
ACM International Conference on the Foundations of Software Engineering (**FSE**) 2024. Link.
Foundation Models for Decision Making (**FMDM**) Workshop at **NeurIPS** 2023

Policy-Guided Lazy Search with Feedback for Task and Motion Planning

Mohamed Khodeir, ***Atharv Sonwane****, Ruthrash Hari, Florian Shkurti*
International Conference on Robotics and Automation (**ICRA**). 2023. Link.
Best Paper Award at the Long-Horizon Planning Workshop, **CoRL** 2022

Neural Analogical Reasoning

Atharv Sonwane*, Abhinav Lalwani, Sweta Mahajan, Gautam Shroff, Lovekesh Vig*
International Workshop on Neural-Symbolic Learning and Reasoning (**NeSy**). 2022. Link.

Solving Visual Analogies Using Neural Algorithmic Reasoning

Atharv Sonwane*, Gautam Shroff, Lovekesh Vig, Ashwin Srinivasan, Tirtharaj Dash.*
Finalist in the Oral Presentation Competition. Student Abstract and Poster Program, **AAAI-22**. Link.

Using Program Synthesis and Inductive Logic Programming to solve Bongard Problems.

Atharv Sonwane*, Sharad Chitlangia*, Tirtharaj Dash, Lovekesh Vig, Gautam Shroff, Ashwin Srinivasan.*
International Workshop on Approaches and Applications of Inductive Programming (**AAIP**) 2021. Link.

Selected Engineering Projects

AutoFill.jl | Julia library for data manipulation | CODE
Implemented FlashFill program synthesis for tabular data analysis during Google Summer of Code (GSoC) 2022

GenRL | PyTorch Reinforcement Learning Library | CODE
Contributed implementations of Deep Contextual Bandits along with distributed RL support using RPC.

Trotbot | Autonomous Delivery Robot | CODE
Obstacle detection and path planning stack built with Robot Operating System (ROS).

Other Roles

Teaching

Cornell	Foundations of AI Reasoning and Decision Making (CS3700)
BITS Goa	Meta Learning (BITS G513), Deep Learning (CS F425), Machine Learning (BITS F464) Discrete Structures for Computer Science (CS F222)

Leadership

- Hardware Lead @ Curem Biotech
- President of the Society for Artificial Intelligence and Deep Learning
- Student Coordinator of the Electronics and Robotics Club

Reviewing

NeurIPS 2024, FMDM Workshop @ NeurIPS 2023

Technical Skills

Programming	Python, Julia, C/C++, MATLAB, SQL, Bash
Deep Learning	PyTorch, NumPy, JAX, pandas, scikit-learn
Robotics	Robot Operating System (ROS), Gazebo, MAVROS, PX4